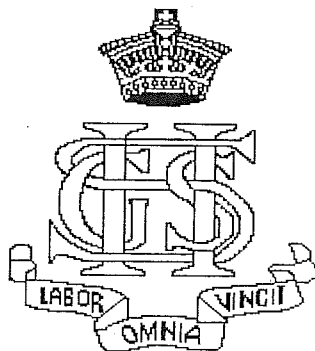


SYDNEY GIRLS HIGH SCHOOL



2006 HSC Assessment Task 1

November 28, 2005

MATHEMATICS Extension 2

Year 12

Time allowed: 90 minutes

Topics: Curve Sketching, Circular Motion

DIRECTIONS TO CANDIDATES:

- Attempt all questions
- Questions are not of equal value
- There are 13 questions
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used
- Write on one side of the paper only
- Use $g = 10 \text{ ms}^{-2}$ in all questions requiring a numerical value of g .

Question 1.

MARKS

Sketch the following curves, showing any important features

$$a) y = \log_3 x, \quad b) x = \sqrt{4 - y^2}, \quad c) y = -|x - 2| \quad [6]$$

Question 2.

Sketch the following polynomials

$$a) y = (x - 1)(x - 2)(x - 3)^2, \quad (b) y = (1 - x)(x - 2)^2(x - 3)^2 \quad [6]$$

Question 3.

Sketch the following curves showing any asymptotes whether they be vertical, horizontal or inclined.

$$a) (x - 2)(y + 3) = 1, \quad b) x^2 - y^2 = 4, \quad c) y = \frac{1}{(x - 1)^2}, \quad d) y = \frac{2x^2 - 3x}{x - 1} \quad [8]$$

Question 4.

Sketch the following piecemeal function showing any points of discontinuity

$$y = \begin{cases} (x + 1)^2 & \text{for } x \leq 0 \\ \sqrt{1 - x^2} & \text{for } 0 < x < 1 \\ 2^{x-1} & \text{for } x \geq 1 \end{cases} \quad [4]$$

$x^2 + y^2 = 1$
 $x = \frac{1}{2}, y = 1$

Question 5.

Find the centre and then sketch the following curves:

$$a) x^2 + y^2 - 8x + 6y = 0, \quad b) x^2 + 4y^2 + 2x - 3 = 0 \quad [6]$$

Question 6.

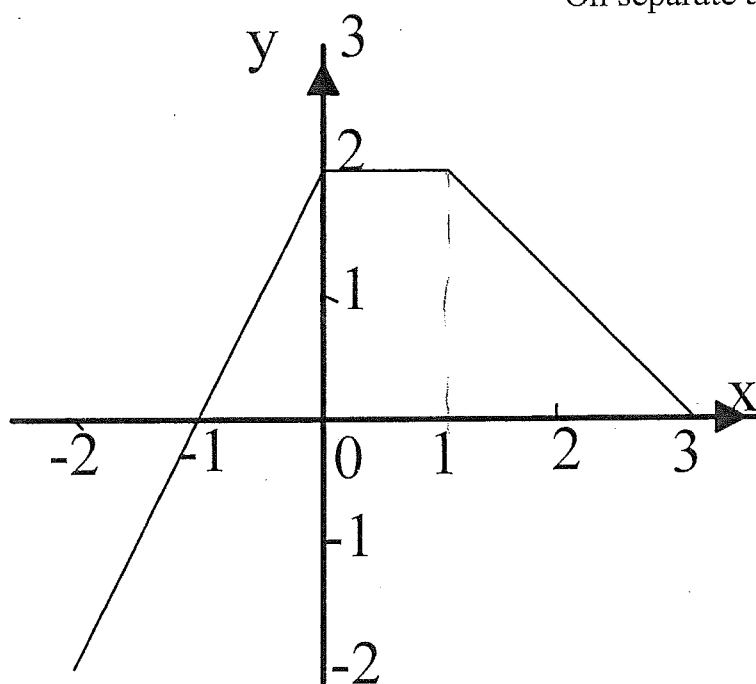
- Sketch the curve $f(x) = 4x - x^2$ on axes with the same horizontal and vertical scale
- Determine the largest possible domain including $x = 3$ for which $f(x) = 4x - x^2$ has an inverse function
- Find the equation of the inverse function $f^{-1}(x)$
- State any common points between $f^{-1}(x)$ and $f(x)$

[8]

Question 7.

The function $y = f(x)$ is shown

On separate axes sketch:



a) $y = \{f(x)\}^2$

b) $y = \frac{1}{f(x)}$

c) $y = f|x|$

d) $y = \sqrt{f(x)}$

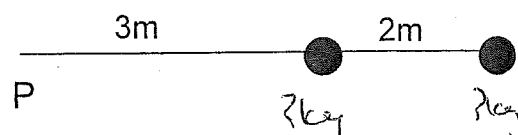
e) $y = 2^{f(x)}$

$y^2 = f(x)$

[10]

Question 8.

A 5 metre long piece of string with two 3 kg masses attached is swung in a horizontal plane around a point P. If the masses are attached 3 metres from P and at the end of the string, what is the maximum angular speed (in radians per second) that the string can be rotated if it has a breaking strain of 600 Newtons



[6]

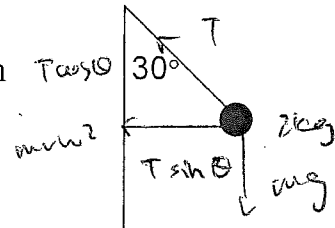
Question 9.

A particle of mass 2 kg is rotating in a conical pendulum with Angle at the vertex 30° . If the particle rotates at 20 rpm, find

- the tension in the string
- the radius of motion

$$20 \times \frac{2\pi}{60}$$

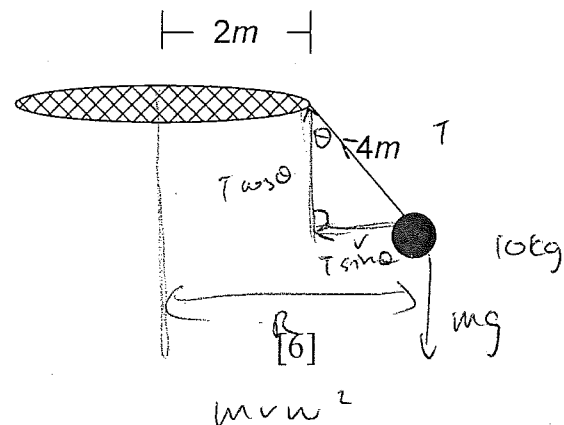
$$\omega = \frac{2\pi}{3}$$



[6]

Question 10.

A 4 metre piece of string is attached to the edge of a disc of radius 2 metres. If a 10 kg mass is attached to the other end of the string, how fast must the system rotate (in radians/second) if the string is to be angled at 30° to the vertical



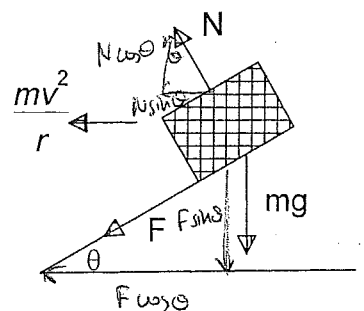
[6]

Question 11.

A road is banked at an angle θ as shown in the adjacent diagram.

- If the force of circular motion is given as $F = \frac{mv^2}{r}$, and the force due to gravity as mg , resolve the frictional force F and reaction force N in the horizontal and vertical directions and hence find an expression for F

- A bus, turning a corner causes the same frictional force down the slope when it travels at 90 km/hr as it does up the slope at 18 km/hr. If the radius of the corner is 260 metres, find (to the nearest degree) the angle at which the road is banked.



fast

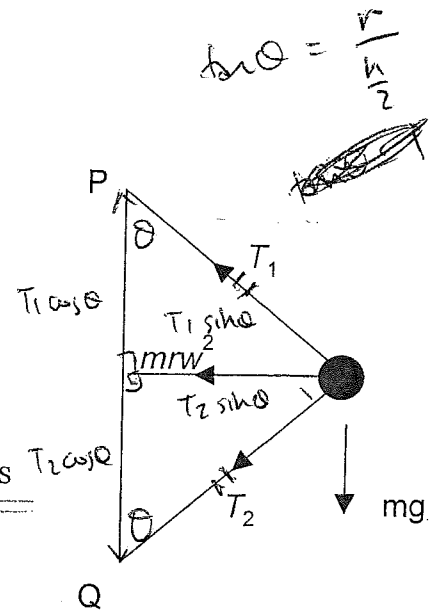
[12]

Question 12

A particle is attached by means of two equal strings to P and Q in the same vertical line. The particle describes a horizontal circle with constant angular speed ω .

a) Prove that in order for the strings to remain taut, $\omega > \sqrt{\frac{2g}{h}}$ where h is the distance PQ

b) If $\omega = 2\sqrt{\frac{2g}{h}}$, find the ratio of the tensions in the two strings



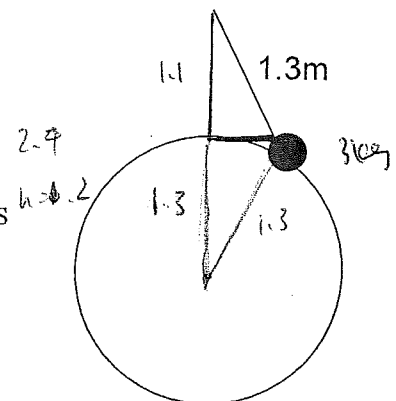
[12]

Question 13.

From a point 1.1 metres above the top of a sphere a 1.3 metre length of string has a 3kg mass attached and that mass is rotating around the surface of the sphere at 2 rad/sec.

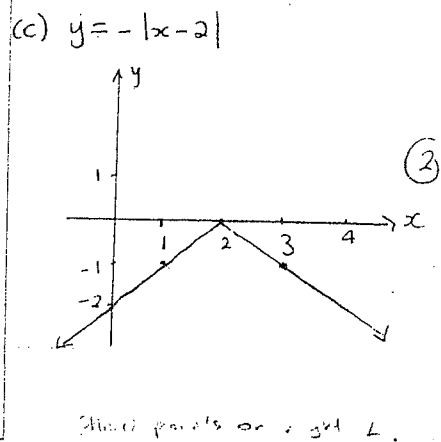
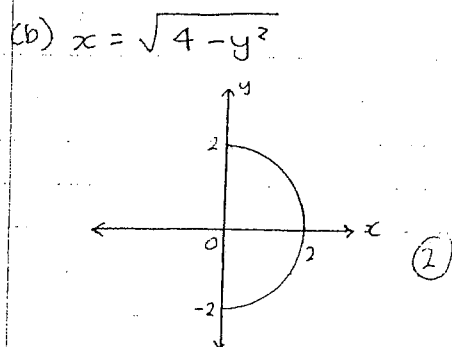
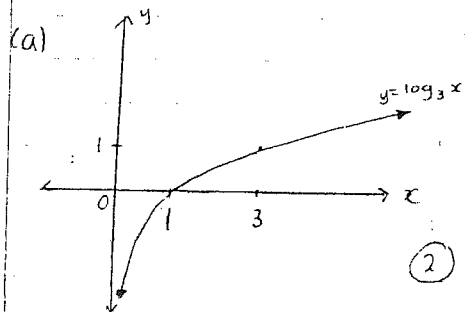
a) Copy the adjacent diagram and show where the forces of tension (T), normal (N), gravity (mg) and circular motion ($mr\omega^2$) lie,

b) If the sphere has a radius (also) of 1.3 metres, determine the force(N) exerted by the sphere on the mass



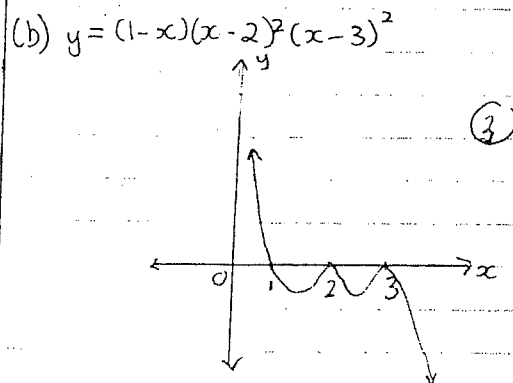
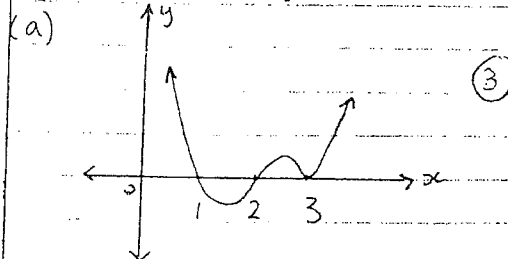
[10]

QUESTION 1 (6 marks)



QUESTION 2 (6 marks)

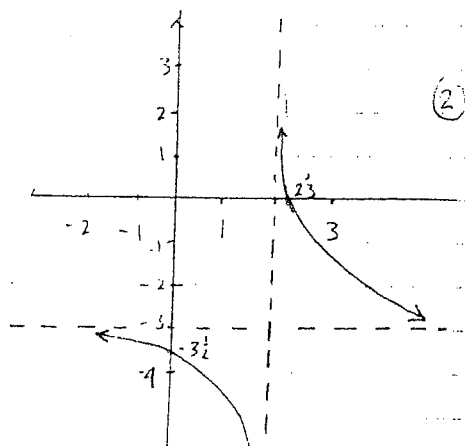
$y = (x-1)(x-2)(x-3)^2$



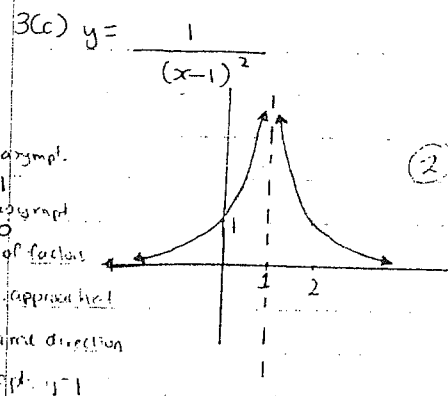
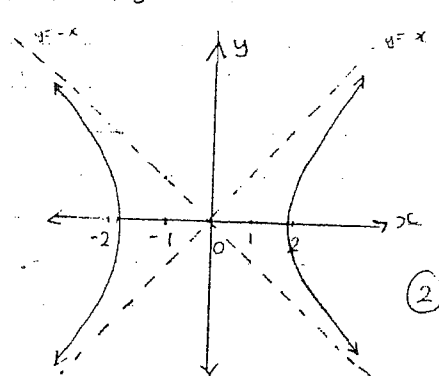
QUESTION 3 (8 marks)

(a) $(x-2)(y+3) = 1$

Vertical asymptote: $x = 2$
Horizontal asymptote: $y = -3$



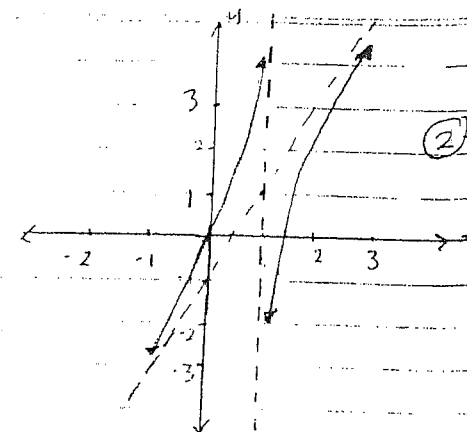
3(b) $x^2 - y^2 = 4$



(d) $y = \frac{2x^2 - 3x}{x-1}$

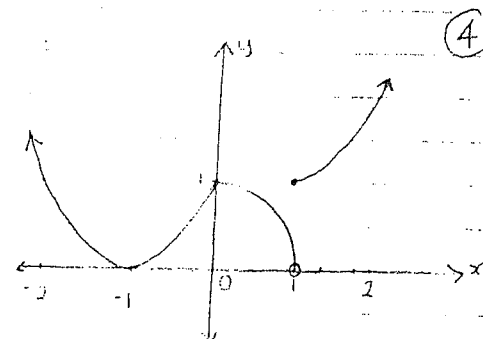
$$\begin{array}{r} 2x^2 - 3x \\ x-1 \overline{) 2x^2 - 3x} \\ \underline{-2x^2 - 2x} \\ -1x + 0 \\ \underline{-1x + 1} \\ -1 \end{array}$$

$y = (2x-1) - \frac{1}{x-1}$
oblique asympt. $y = 2x-1$
Vertical asympt. $x = 1$



QUESTION 4 (4 marks)

$y = \begin{cases} (x+1)^2 & \text{for } x \leq 0 \\ \sqrt{1-x^2} & \text{for } 0 < x < 1 \\ 2^{x-1} & \text{for } x \geq 1 \end{cases}$

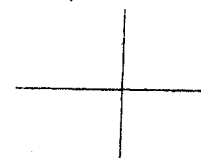


QUESTION 5 (6 marks)

(a) $x^2 + y^2 - 8x + 6y = 0$

$x^2 - 8x + (-4)^2 + y^2 + 6y + (-3)^2 = 16 + 9$
 $(x-4)^2 + (y+3)^2 = 25$

circle with centre (4, -3)
and radius = 5 units

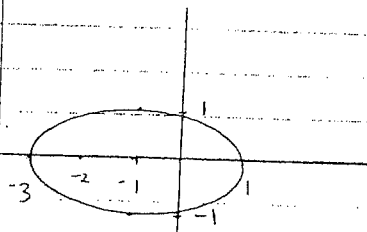


Q5(b) $x^2 + 2x + (1)^2 + 4y^2 = 3 + 1^2$

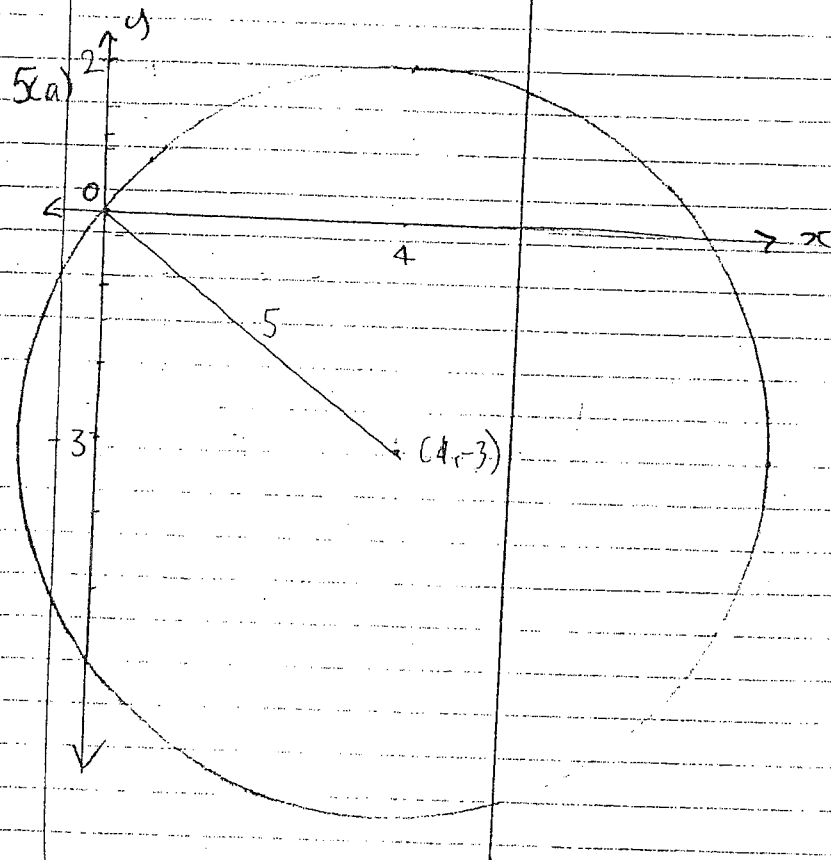
$(x+1)^2 + 4y^2 = 4$

$\frac{(x+1)^2}{4} + y^2 = 1$

Ellipse with centre $(-1, 0)$

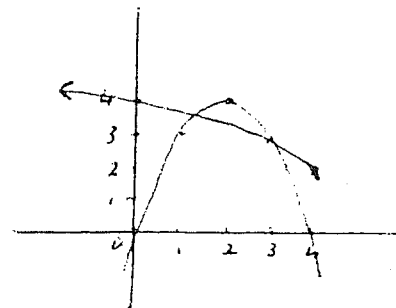


(3)



Q6

a)



b) Turning pt at $(2, 4)$
 $\therefore$ Domain is $x \geq 2$

c) Let $x = 4y - y^2$

$y^2 - 4y + 4 = 4 - x$

$(y-2)^2 = 4-x$

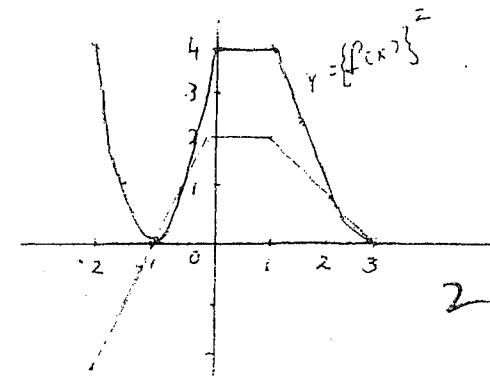
$y-2 = \pm \sqrt{4-x}$

$y-2 = \pm \sqrt{4-x}$ if originally $x \geq 3$

$y = 2 + \sqrt{4-x}$

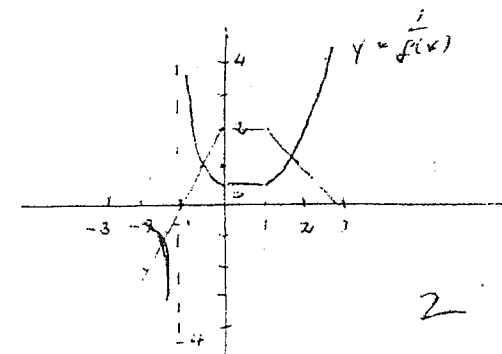
d) The common point is $(3, 3)$
 (from the diagram)

2



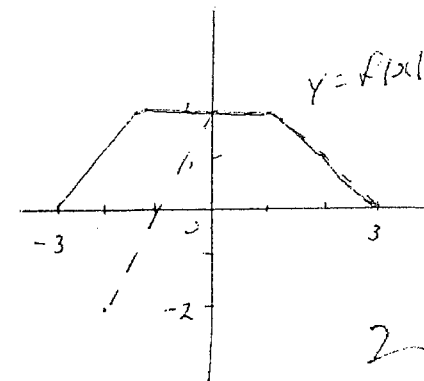
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b)

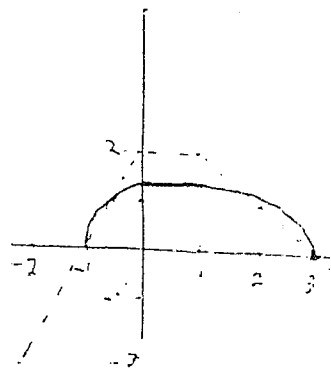


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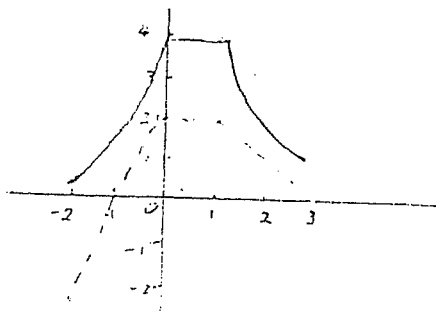
c)



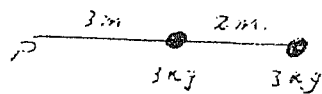
2



e) $y = 2^{f(x)}$



RP,



$$T = m_1 \omega^2 + m_2 \omega^2$$

$$600 = 3 \times 3 \times \omega^2 + 3 \times 5 \times \omega^2$$

$$600 = 9\omega^2 + 15\omega^2$$

$$600 = 24\omega^2$$

$$25 = \omega^2$$

$$\therefore \omega = 5 \text{ rad/sec.}$$

6

Q9

$$\omega = 20 \text{ rpm}$$

$$= \frac{20 \times 2\pi}{60}$$

$$= 2\pi/3 \text{ rad/sec.}$$

$$a) T \cos 30 = mg$$

$$\therefore T \cdot \frac{\sqrt{3}}{2} = 2 \times 10$$

$$\therefore T = 40/\sqrt{3} \text{ N}$$

$$b) T \sin 30 = m r \omega^2$$

$$\therefore \frac{40}{\sqrt{3}} \cdot \frac{1}{2} = 2 \times \pi \times \left(\frac{2\pi}{3}\right)^2$$

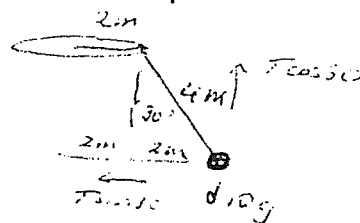
$$20/\sqrt{3} = \frac{8\pi^2}{9} r$$

$$\therefore \frac{180}{8\sqrt{3}\pi^2} = r$$

$$r = \frac{45}{2\sqrt{3}\pi^2} \text{ m.} = \frac{15\sqrt{3}}{2\pi^2} \text{ m.}$$

$$\approx 1.32 \text{ m}$$

Q10



$$T \cos 30 = mg$$

$$T \sin 30 = m R \omega^2$$

$$\text{where } R = 2 + 4 \sin 30 = 4$$

$$\therefore \tan 30 = \frac{R \omega^2}{g}$$

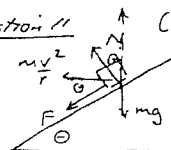
$$\frac{10 \times 1}{4 \sqrt{3}} = \omega^2$$

$$\therefore \omega = \sqrt{\frac{10}{4\sqrt{3}}}$$

$$= \sqrt{\frac{5}{2\sqrt{3}}} \text{ rad/sec.}$$

$$\approx 1.2 \text{ rad/sec.}$$

Question 11 (12 marks)



a) Vertically

$$1) N \cos \theta = F \sin \theta + mg \quad - (1) \times \sin \theta$$

Horizontally

$$1) N \sin \theta + F \cos \theta = m \frac{v^2}{r} \quad - (2) \times \cos \theta$$

$$N \sin \theta \cos \theta + F \sin \theta + mg \sin \theta = m \frac{v^2}{r} \cos \theta$$

$$N \sin \theta \cos \theta + F \cos \theta = m \frac{v^2}{r} \cos \theta$$

$$\therefore F \sin \theta + mg \sin \theta + F \cos \theta = m \frac{v^2}{r} \cos \theta$$

$$3) F = m \frac{v^2}{r} \cos \theta - mg \sin \theta \quad *$$

$$b) 90 \text{ km/hr} \Rightarrow \frac{90,000}{3600} \text{ m/s} = 25 \text{ m/s}$$

$$18 \text{ km/hr} \Rightarrow \frac{18,000}{3600} \text{ m/s} = 5 \text{ m/s}$$

$$r = 260 \text{ m}$$

$$\text{Now } F_1 = -F_2$$

$$\therefore m \frac{v_1^2}{r} \cos \theta - mg \sin \theta = - \left(m \frac{v_2^2}{r} \cos \theta - mg \sin \theta \right)$$

$$\therefore m \left(\frac{v_1^2}{r} \cos \theta + \frac{v_2^2}{r} \cos \theta \right) = 2mg \sin \theta$$

$$\left(\frac{25^2}{260} \cos \theta + \frac{5^2}{260} \cos \theta \right) = 20 \sin \theta$$

$$\frac{650}{260} \cos \theta = 20 \sin \theta$$

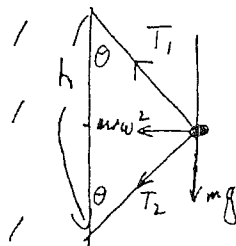
$$\frac{650}{20 \times 260} = \tan \theta$$

$$\theta \approx 7^\circ$$

5

Question 12 (12 marks)

3



Vertically:

$$T_1 \cos \theta = T_2 \cos \theta + mg.$$

Horizontally:

$$T_1 \sin \theta + T_2 \sin \theta = m r \omega^2.$$

For strings to remain taut

$$T_2 > 0$$

$$\text{Take } T_2 = 0$$

$$\therefore T_1 \cos \theta = mg.$$

$$T_1 \sin \theta = m r \omega^2.$$

$$\therefore \tan \theta = \frac{r \omega^2}{g}.$$

$$\tan \theta = \frac{r}{h}.$$

$$\Rightarrow \tan \theta = \frac{2r}{h}$$

$$\therefore \frac{2r}{h} = \frac{r \omega^2}{g}.$$

$$\therefore \omega^2 = \frac{2g}{h} \Rightarrow \omega = \sqrt{\frac{2g}{h}}.$$

3 Now if $T_2 > 0$ then $\omega > \sqrt{\frac{2g}{h}}$

$$\omega = 2 \sqrt{\frac{2g}{h}}$$

horizontally.

$$T_1 \sin \theta + T_2 \sin \theta = m r \times 4 \left(\frac{2g}{h} \right)$$

$$\text{Now } \tan \theta = \frac{2r}{h} \quad T_1 \sin \theta + T_2 \sin \theta = m \tan \theta \times 4g.$$

$$T_1 \sin \theta + T_2 \sin \theta = m \frac{2r}{h} \times 4g.$$

$$\therefore T_1 \cos \theta + T_2 \cos \theta = 4mg.$$

$$\therefore T_1 \cos \theta + mg + T_2 \cos \theta = 4mg.$$

$$2T_2 \cos \theta = 3mg.$$

$$T_2 \cos \theta = \frac{3}{2} mg.$$

$$T_1 \cos \theta = \frac{3}{2} mg + mg$$

$$T_1 \cos \theta = \frac{5}{2} mg.$$

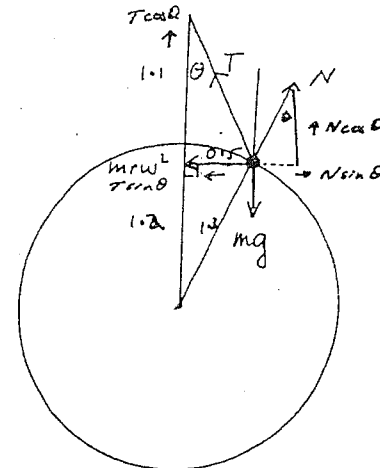
$$\text{Now } T_1 \cos \theta : T_2 \cos \theta = \frac{5}{2} mg : \frac{3}{2} mg.$$

4

$$T_1 : T_2 = 5 : 3$$

Question 13 (10 marks)

2



$$m = 3 \text{ kg}$$

$$\omega = 2 \text{ rad/sec}$$

$$r = \frac{1}{2} m.$$

$$\sin \theta = \frac{5}{13}$$

$$\cos \theta = \frac{12}{13}$$

5 Vertically:

$$T \cos \theta + N \cos \theta = mg.$$

$$2 \quad \frac{12}{13} T + \frac{12}{13} N = 30$$

$$12T + 12N = 30 \times 13 \Rightarrow 12T = 30 \times 13 - 12N.$$

$$T = \frac{1}{12} \{ 30 \times 13 - 12N \}$$

Horizontally:

$$T \sin \theta - N \sin \theta = m r \omega^2$$

$$2 \quad \frac{5}{13} T - \frac{5}{13} N = 3 \times \frac{1}{2} \times 4$$

$$5T - 5N = 6 \times 13 \Rightarrow 5T = 5N + 6 \times 13$$

$$T = \frac{1}{5} \{ 5N + 6 \times 13 \}$$

$$\therefore \frac{1}{12} \{ 30 \times 13 - 12N \} = \frac{1}{5} \{ 5N + 6 \times 13 \}.$$

$$5 \{ 30 \times 13 - 12N \} = 12 \{ 5N + 6 \times 13 \}.$$

$$1950 - 60N = 60N + 936$$

4

$$120N = 1014$$

$$N = \underline{8.45 \text{ Newtons}}$$